THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Final Subject Assessment, 2023

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 28 pages (including this page) with 11 questions and 90 total marks

Permitted Materials

- One double-sided handwritten A4 sheet
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. If you do not have enough space, tick the box near the bottom of page 28 and request a booklet from an invigilator—include the question number at the top of each page in the booklet.
- You must NOT remove this question paper, or any booklets provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request a booklet if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with answer booklets if any, at the conclusion of the examination.

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Question 1 (6 marks)

(a) Let p and q be primitive statements. Write a truth table for the compound statement

$$[q \vee (p \wedge (\sim q))] \implies q$$

Is this statement a tautology? Explain.

(b) Let (a_n) be a sequence of real numbers. Write the following statement in formal mathematical language:

(a_n) *is neither monotone increasing nor monotone decreasing.*

(a)

p	q	$\sim q$	$p \wedge (\sim q)$	$q \vee (p \wedge (\sim q))$	$[q \vee (p \wedge (\sim q))] \implies q$
T	T	F	F	T	T
T	F	T	T	T	F
F	T	F	F	T	T
F	F	T	F	F	T

The statement $[q \vee (p \wedge (\sim q))] \implies q$ is false when p is true and q is false; therefore, it is not a tautology, since it is not true regardless of the truth value of the primitive statements.

(b) **Method 1**

$$[(\exists n \in \mathbb{N}^+)\ (a_n > a_{n+1})] \wedge [(\exists n \in \mathbb{N}^+)\ (a_n < a_{n+1})]$$

Method 2

$$\sim \Big[\big((\forall n \in \mathbb{N}^+)\ (a_n < a_{n+1})\big) \vee \big((\forall n \in \mathbb{N}^+)\ (a_n > a_{n+1})\big)\Big]$$

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Question 2 (7 marks)

Using a proof by induction, prove that $2^n \leq (n+1)!$ for all $n \in \mathbb{N}$.

Proof. We proceed by induction on $n \geq 0$. Let $p(n)$ be the statement

$$2^n \leq n!$$

1. *Show that $p(0)$ is true.*
   When $n = 0$, $2^n = 2^0 = 1 \leq 1 = 1! = (0+1)!$, so $p(0)$ is true.
2. *Show that for each $k \in \mathbb{N}$, if $p(k)$ is true then $p(k+1)$ is true.*
   Suppose that for some $k \in \mathbb{N}$, $p(k)$ is true.
   In other words, $2^k \leq (k+1)!$.
   Then

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &\leq 2 \cdot (k+1)! \\ &\leq (k+2) \cdot (k+1)! \\ &= (k+2)! \end{aligned}$$

   Here the first $\leq$ holds by the inductive hypothesis that $p(k)$ is true, and the second because $2 \leq k+2$ as $k \in \mathbb{N}$.
   Thus, $2^{k+1} \leq ((k+1)+1)!$, that is, $p(k+1)$ holds true.

Therefore, by the Principle of Mathematical Induction, $p(n)$ holds true for all $n \in \mathbb{N}$. That is, $2^n \leq (n+1)!$ for all $n \in \mathbb{N}$.

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Question 3 (7 marks)

Let U be a set. Let A, B, and C be subsets of U.

(a) Write a (semi-)formal proof of the following statement:

$$(A \setminus B) \cup (A \setminus C) \subseteq A \setminus (B \cap C)$$

(b) Is it true that

$$(A \setminus B) \cup (A \setminus C) = A \setminus (B \cap C)\ ?$$

Justify your answer using a (semi-)formal proof or a specific counter-example.

(a) **Proof.**

1. Let A, B, and C be subsets of a set U.	(premise)
2. Let $x \in (A \setminus B) \cup (A \setminus C)$.	(premise)
3. $(x \in A \setminus B) \vee (x \in A \setminus C)$	(2, defn of $\cup$)
4. $x \in A \setminus B$	(3, premise)
5. $x \in A$	(4., defn of $\setminus$)
6. $x \notin B$	(4, defn of $\setminus$)
7. $x \notin B \cap C$	(6, defn of $\cap$)
8. $x \in A \setminus (B \cap C)$	(4, 7, defn of $\setminus$)
9. $x \in A \setminus C$	(3, premise)
10. $x \in A$	(9., defn of $\setminus$)
11. $x \notin C$	(9, defn of $\setminus$)
12. $x \notin B \cap C$	(11, defn of $\cap$)
13. $x \in A \setminus (B \cap C)$	(10, 12, defn of $\setminus$)
14. $(A \setminus B) \cup (A \setminus C) \subseteq A \setminus (B \cap C)$.	(3, 8, 13)

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(b) The statement is true. To show this, we need to prove the reverse inclusion

$$A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C)$$

Proof.

1. Let A, B, and C are subsets of a set U.	(premise)
2. Let $x \in A \setminus (B \cap C)$.	(premise)
3. $x \in A$	(2, defn of $\setminus$)
4. $\sim [(x \in B) \wedge (x \in C)]$	(2, defn of $\setminus$, $\cap$)
5. $(x \notin B) \vee (x \notin C)$	(4)
6. $(x \in A \setminus B) \vee (x \in A \setminus C)$	(3, 5, defn of $\setminus$)
7. $x \in (A \setminus B) \cup (A \setminus C)$	(6, defn of $\cup$)
8. $A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C)$.	(2, 7)

Question 4 (5 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists 1/x \in \mathbb{R} \quad x \cdot 1/x = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Let $A \subset \mathbb{R}$ be a bounded above subset, and c an upper bound for A. Define a subset $-A = \{x \in \mathbb{R} | -x \in A\}$. Using only the Axioms of the Real Numbers (and rules of logic), write a (semi-)formal proof showing that $-c$ is a lower bound for $-A$.

Proof.

1. $(\forall y \in A)\ y \leq c$	(premise)
2. $x \in -A.$	(premise)
3. $-x \in A$	(defn of $-A$)
4. $-x \leq c$	(3, hypothesis on c)
5. $-x + x \leq c + x$	(4, OA)
6. $x + (-x) \leq c + x$	(5, A2)
7. $0 \leq c + x$	(6, A5)
8. $0 \leq x + c$	(7, A2)
9. $0 + (-c) \leq (x + c) + (-c)$	(8, OA)
10. $-c + 0 \leq (x + c) + (-c)$	(9, A2)
11. $-c \leq (x + c) + (-c)$	(10, A4)
12. $-c \leq x + (c + (-c))$	(11, A3)
13. $-c \leq x + 0$	(12, A5)
14. $-c \leq x$	(13, A4)
15. $(\forall x \in -A)\ -c \leq x$	(2, 14)

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Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 5 (9 marks)

In each part of this question, you are asked to give an example of something, and explain why it is an example.

(a) A subset $A \subseteq \mathbb{R}$ that is neither bounded below nor bounded above and has no limit points.

(b) A subset $A \subseteq \mathbb{R}$ consisting only of rational numbers, but so that the supremum $\sup(A)$ is irrational. (You do not need to prove that any of the "usual" irrational numbers are irrational.)

(c) A non-empty bounded subset $A \subseteq \mathbb{R}$ such that every $a \in A$ is a limit point of A.

(a) The set of integers $\mathbb{Z}$ is an example. It is unbounded by the Archimedean principle. Take $a \in \mathbb{R}$. If $a \notin \mathbb{Z}$, then there exists an integer n such that $n < x < n+1$; let $\delta = \min\{|x-n|, |n+1-x|\}$. Then the interval $(a-\delta, a+\delta)$ contains no integer, so a is not a limit point of $\mathbb{Z}$. If $a \in \mathbb{Z}$, take $\delta = 1$. Then the interval $(a-\delta, a+\delta)$ contains no integers aside from a itself, so again, a is not a limit point of $\mathbb{Z}$.

(b) Take any irrational number, for example $\sqrt{2}$. Let $A = \{x \in \mathbb{Q} | x < \sqrt{2}\}$. I claim $\sup(A) = \sqrt{2}$. It is an upper bound by construction. Suppose $L < \sqrt{2}$. Then, by density of the rationals, there is some rational number x such that $L < x < \sqrt{2}$. By definition, $x \in A$; it follows that L is not an upper bound of A. Therefore, $\sqrt{2}$ is the least upper bound.

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(c) Any interval, open, closed, half open, will work. For example, take $A = (0, 1)$. Let $a \in (0, 1)$, and $\delta > 0$. Let $c = \max\{0, a - \delta\}$; since $a > c$, there is a real number x with $c < x < a$. Because $c \geq 0$, $x \in (0, 1)$, and because $c \geq a - \delta$, $|a - x| < \delta$. Therefore, a is a limit point of $(0, 1)$.
(For a closed interal, a separate, but similar, argument is needed for the end points.)

Question 6 (9 marks)

Consider the sequence (f_n) defined by

$$f_n = \frac{2n+1}{5n+2}$$

for each $n \in \mathbb{N}^+$.

(a) Show that (f_n) is monotone decreasing and bounded below.

(b) Use the ϵ - M definition of convergence to show that (f_n) converges to $2/5$.

(c) Find the infimum of the set $\{f_n | \, n \in \mathbb{N}^+\}$. Briefly explain.

(a) We compute

$$f_{n+1} - f_n = \frac{2(n+1)+1}{5(n+1)+2} - \frac{2n+1}{5n+2} = \frac{-1}{25n^2+45n+14} < 0$$

for all $n \in \mathbb{N}^+$. Therefore, (f_n) is monotone decreasing. We also note that, for all $n \in \mathbb{N}^+$, $f_n > 0$. Thus, (f_n) is bounded below.

(b) **Scrap work.** Let $\epsilon > 0$. Then

$$\begin{aligned} \left|\frac{2n+1}{5n+2} - \frac{2}{5}\right| &< \epsilon \\ \left|\frac{1}{5(5n+2)}\right| &< \epsilon \\ \frac{1}{5\epsilon} &< 5n+2 \\ \frac{1}{25\epsilon} - \frac{2}{5} &< n \end{aligned}$$

Let $M = \max\{1, \lceil \frac{1}{25\epsilon} - \frac{2}{5} \rceil\}$. (Note: instead, we can just choose $M = \lceil \frac{1}{25\epsilon} \rceil$.)

Proof. Let $\epsilon > 0$ and let $M = \max\{1, \lceil \frac{1}{25\epsilon} - \frac{2}{5} \rceil\}$. For each $n > M$

$$\begin{aligned} \frac{1}{25\epsilon} - \frac{2}{5} &< n \\ \frac{1}{5\epsilon} &< 5n+2 \end{aligned}$$

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$$\frac{1}{5(5n+2)} < \epsilon$$
$$\left|\frac{1}{5(5n+2)}\right| < \epsilon$$
$$\left|\frac{2n+1}{5n+2} - \frac{2}{5}\right| < \epsilon.$$

Therefore, if $n > M$ then $|f_n - 2/5| < \epsilon$. That is, (f_n) converges to $2/5$.

(c) The infimum is $2/5$. We saw in (a) that (f_n) is monotone decreasing and bounded below; thus it converges to the infimum of its set of terms. But we proved in (b) that $f_n \to 2/5$.

Question 7 (11 marks)

Let (f_n) be a sequence converging to a limit L, and let (a_n) be the sequence defined by $a_n = (-1)^n f_n$ for $n \in \mathbb{N}^+$.

(a) Assume $L = 0$. Using the ϵ - M definition of convergence, show that a_n also converges to 0.

(b) Now assume $L \neq 0$. Using theorems from lectures, show that the sequence (a_n) has two subsequences converging to different limits.

(c) If $L \neq 0$, is (a_n) a Cauchy sequence? Briefly explain.

(a) Let $\epsilon > 0$. Since $f_n \to 0$, there exists $M \in \mathbb{N}$ such that, for all $n > M$, $|f_n| = |f_n - 0| < \epsilon$. Thus, if $n > M$, then

$$|a_n - 0| = |a_n| = |f_n| < \epsilon$$

It follows that $a_n \to 0$.

(b) Consider the two subsequences (a_{2n}) and (a_{2n+1}). We have $a_{2n} = f_{2n}$ and $a_{2n+1} = -f_{2n+1}$. Since f_n converges, all its subsequences also converge, to the same limit L. Thus, $f_{2n} \to L$ and $f_{2n+1} \to L$. Applying the Algebra of Limits Theorem, we conclude that $a_{2n} \to L$ and $a_{2n+1} \to -L$. Finally, since $L \neq 0$, $L \neq -L$.

(c) Since (a_n) has two subsequences converging to different limits, it does not converge. It follows that it is not a Cauchy sequence.

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Question 8 (9 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(a) Using continuity of the sine function, and the ϵ - M definition of limits, prove that $\lim_{x\to\infty} f(x) = 0$.

(b) Using the ϵ - δ definition of limits, show that f does not converge as x approaches 0.

(c) Is f continuous at 0? Briefly explain your answer.

(a) Let $\epsilon > 0$. Since the sine function is continuous, there exists $\delta > 0$ such that, whenever $|z| < \delta$, $|\sin(z)| < \epsilon$. By the Archimedean principle, there exists $M > 0$ such that, for all $x > M$, $0 < \frac{1}{x} < \delta$. Combining these two, it follows that for all $x > M$, $|f(x)| = \left|\sin(\frac{1}{x})\right| < \epsilon$. By the ϵ - M definition of the limit as x approaches ∞, $\lim_{x\to\infty} f(x) = 0$.

(b) Suppose for a contradiction that there exists $L \in \mathbb{R}$ such that $\lim_{x\to 0} f(x) = L$. Let $\epsilon = 1/2$. Then there exists $\delta > 0$ such that, whenever $0 < x < \delta$, then $|f(x) - L| < 1/2$. By the Archimedean principle, there is a natural number n such that $x_+ = 1/(2n\pi + \frac{\pi}{2}) < \delta$. Thus,

$$\left|\sin(\frac{1}{x_+}) - L\right| = \left|\sin(2n\pi + \frac{\pi}{2}) - L\right| = |1 - L| < 1/2.$$

It follows that $L > 1/2$. On the other hand, for the same n, $x_- = 1/((2n+1)\pi + \frac{\pi}{2}) < \delta$ as well. Therefore,

$$\left|\sin(\frac{1}{x_-}) - L\right| = \left|\sin((2n+1)\pi + \frac{\pi}{2}) - L\right| = |-1 - L| < 1/2.$$

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It follows that $L < -1/2$. This is a contradiction. Therefore, $\lim_{x \to 0} f(x) \neq L$, for all $L \in \mathbb{R}$. That is, f diverges as x approaches 0.

(c) f is not continuous at 0. Since 0 is a limit point of the domain, f is continuous if, and only if, $\lim_{x \to 0} f(x) = f(0)$. But we saw in (b) that this limit does not converge.

Question 9 (6 marks)

Let $f : [1,3] \to \mathbb{R}$ be defined by

$$f(x) = \frac{1}{\sqrt{x^3 + x}}$$

Without computing the integral explicitly, show that

$$\frac{\sqrt{30} + 3\sqrt{10}}{30} \leq \int_1^3 \frac{1}{\sqrt{x^3 + x}}\, dx \leq \frac{\sqrt{10} + 5\sqrt{2}}{10}.$$

(Hint: Consider the partition $\{1, 2, 3\}$ of $[1, 3]$. You may use without proof that the function $\sqrt{x^3 + x}$ is increasing on its domain $[0, \infty)$.)

Notice that

$$\sqrt{x^3 + x}$$

is increasing on the interval $[1, 3]$, so that $f(x)$ is decreasing. Thus, for any partition, the lower Riemann sum can be computed by evaluating the f at the end points of subintervals, and the upper Riemann sum by evaluating at the starting points.

We compute that:

$$f(1) = \frac{1}{\sqrt{2}}$$
$$f(2) = \frac{1}{\sqrt{10}}$$
$$f(3) = \frac{1}{\sqrt{30}}$$

Hence the lower Riemann sum with respect to the partition $P = \{1, 2, 3\}$ is

$$\begin{aligned} L(f, P) &= f(2)(2-1) + f(3)(3-2) \\ &= \frac{1}{\sqrt{10}} + \frac{1}{\sqrt{30}} \\ &= \frac{\sqrt{30} + 3\sqrt{10}}{30}. \end{aligned}$$

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And the upper Riemann sum with respect to the partition $P = \{1, 2, 3\}$ is

$$\begin{aligned} U(f, P) &= f(1)(2-1) + f(2)(3-2) \\ &= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{10}} \\ &= \frac{\sqrt{10} + 5\sqrt{2}}{10}. \end{aligned}$$

Therefore, by the definition of the Riemann integral of f

$$\frac{\sqrt{30} + 3\sqrt{10}}{30} = L(f, P) \leq \int_1^3 \frac{1}{\sqrt{x^3 + x}}\, dx \leq U(f, P) = \frac{\sqrt{10} + 5\sqrt{2}}{10}.$$

Note. *For Questions 10 and 11, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 10 (12 marks)

(a) For each of the following infinite series, use an appropriate test or theorem to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{2n^2+2}$

(ii) $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{n^3}$

(b) Find the radius of convergence and the interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{x^n}{3^n\, n}.$$

(a) (i) Diverges.

If a series $\sum a_n$ converges, then the sequence of terms $a_n \to 0$, and therefore the sequence $|a_n| \to 0$ as well. For this series, the sequence $|a_n| = \frac{n^2}{2n^2+2} \to 1/2$ by the algebra of limits theorem. Thus, the series diverges. (We can also use Question 7 to show that the series of terms diverges, but that is a more complicated argument.)

Answer: Series diverges.
Correct justification.

(ii) Converges.

By the p-series test with $p = 2$, the series $\displaystyle\sum_{n=1}^{\infty} \frac{2}{n^2}$ converges (absolutely). Since $0 \leq \frac{n+1}{n^3} \leq \frac{2}{n^2}$ for all $n \in \mathbb{N}^+$, our series converges (absolutely) by the comparison test.

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(b) By the Ratio Test, since

$$\begin{aligned} r &= \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} \\ &= \lim_{n\to\infty} \frac{|x|^{n+1}}{3^{n+1}(n+1)} \cdot \frac{3^n\, n}{|x|^n} \\ &= \lim_{n\to\infty} \frac{n\,|x|}{3(n+1)} \\ &= \frac{|x|}{3}, \end{aligned}$$

the power series converges if $r = |x|/3 < 1$, i.e. if $|x| < 3$.

Moreover, the power series diverges if $r = |x|/3 > 1$, i.e. if $|x| > 3$.

In particular, the radius of convergence is 3.

Correct answer for radius of convergence.
Correct use of Ratio test to find the radius of convergence.

At the endpoint $x = -3$, by the Alternating Series Test since $\frac{|x|^n}{3^n\, n} = \frac{1}{n}$ is monotone decreasing and

$$\lim_{n\to\infty} \frac{|x|^n}{3^n\, n} = \lim_{n\to\infty} \frac{1}{n} = 0,$$

the power series converges.

At the endpoint $x = 3$, by the p-Series Test with $p = 1$, the power series diverges.

Therefore, the interval of convergence is $[-3, 3)$.

Question 11 (9 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be a function which is infinitely differentiable, i.e. the n-th derivative $f^{(n)}$ of f exists and is differentiable on $\mathbb{R}$ for all $n \in \mathbb{N}^+$. Assume also that there exists $k \in \mathbb{N}^+$ such that $f^{(k)}(x) = f(x)$.

(a) Show that there exists $\ell > 0$ such that $\left|f^{(n)}(0)\right| \leq \ell$ for all $n \in \mathbb{N}$.

(b) Show that the Taylor series $T_f(x)$ of f at 0 has infinite convergence radius. (You may use that the exponential series has infinite convergence radius.)

(c) Using Taylor's Theorem, prove that $f(x) = T_f(x)$ for all $x \in \mathbb{R}$.

(a) Take $\ell = \max\{|f(0)|, \ldots, \left|f^{(k-1)}(0)\right|\}$. Let $n \in \mathbb{N}$, and write $n = q \cdot k + r$, with $0 \leq r < k$ (division with remainder). Then $\left|f^{(n)}(0)\right| = \left|f^{(r)}(0)\right| \leq \ell$.

(b) From (a), we have for all $x \in \mathbb{R}$,

$$\sum_{n=0}^{\infty} \left|\frac{f^{(n)}(0)}{n!}\right| |x|^n \leq \sum_{n=0}^{\infty} \left|\frac{\ell}{n!}\right| |x|^n = \ell\, e^{|x|}.$$

By the comparison test, $T_f(x)$ converges absolutely for all real numbers x, and therefore has infinite convergence radius.

(c) Let T_n denote the order n Taylor polynomial of f at 0, and let $x \in \mathbb{R}$. For $n \in \mathbb{N}$, let $M_n = \sup\{\left|f^{(n)}(y)\right| \ |y \in [-x, x]\}$. This supremum exists, because all the derivatives of an infinitely differentiable function are continuous. Set $C = \max\{M_0, \ldots, M_{k-1}\}$. Arguing as in part (a), we conclude: if $c \in [-x, x]$ and $n \in \mathbb{N}$, then $\left|f^{(n+1)}(c)\right| \leq C$.

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Now we apply Taylor's Theorem. There exists $c \in [-x, x]$ such that

$$f(x) - T_n(x) = \frac{f^{(n+1)}(c)}{n!} x^{n+1}.$$

Since $|f^{(n+1)}(c)| \leq C$ for all $n \in \mathbb{N}$,

$$|f(x) - T_n(x)| \leq \frac{C}{n!} |x|^{n+1}.$$

Since

$$\lim_{n\to\infty} \frac{C}{n!} |x|^{n+1} = 0,$$

we have that $(T_n(x))$ converges to $f(x)$ as $n \to \infty$.

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End of Exam — Total Available Marks = 90

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